Dense Passage Retrieval and Reader Reranking on a 50K-Passage Wikipedia Corpus

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***Abstract*—Reduced-corpus experiments make dense passage retrieval accessible but change both answer availability and ranking difficulty. We evaluate official Dense Passage Retrieval (DPR) checkpoints on a fixed reservoir sample of 50,000 Wikipedia passages. Full-test references cover five benchmarks, while Reader reranking uses all 15,617 questions from Natural Questions, CuratedTREC, and TriviaQA. Exact FAISS IndexFlatIP supplies 200 candidates to the original Reader relevance head. Answer-containing-passage accuracy at rank 20 increases from 25.68% to 27.34% on Natural Questions, from 16.86% to 18.59% on CuratedTREC, and from 33.60% to 42.64% on TriviaQA. Answer-string corpus coverage is uneven at 60.72%, 31.27%, and 72.77%, respectively. Conditioning on coverage changes the cross-dataset interpretation: CuratedTREC has low absolute accuracy but a competitive hit rate among covered questions. Increasing the candidate budget from 100 to 200 improves Natural Questions accuracy at rank 100 but slightly reduces rank-20 accuracy on both tests with completed budget comparisons, at approximately twice the reranking time. The findings characterize inference with released models, separating corpus coverage, ranking quality, and computation without claiming a new architecture or a reproduction of the original training schedule.**

***Index Terms*—dense passage retrieval, passage reranking, corpus coverage, FAISS, open-domain question answering**

I. Introduction

Open-domain question answering depends on finding useful evidence in a large text collection. A retrieval system may identify a passage that discusses the topic of a question without placing a passage containing its answer near the top. This distinction becomes important when a lightweight first-stage retriever is followed by a more expensive neural reader. A system can lose answer-bearing evidence because the corpus omits it, because retrieval misses it, or because the final ranking places it below the evaluation cutoff.

Dense Passage Retrieval (DPR) provides a dual-encoder implementation and public model and data resources for studying this pipeline [1], [2]. Its original evaluation searches a Wikipedia collection of approximately 21 million passages. Reusing this full setting involves substantial storage, passage encoding, and indexing costs. A smaller collection makes an experimental pipeline more accessible, but also changes the task: it removes many possible answer passages along with many distractors. Results from a reduced collection therefore cannot be read as scores obtained under the original paper's conditions.

Our study fixes a reservoir sample of 50,000 passages and investigates how existing components behave under that constraint. We use exact inner-product search and retain the pretrained DPR weights. We then add a second stage that scores each retrieved question–passage pair with the relevance head of the official extractive DPR Reader. This setup permits a practical investigation of ranking improvements without attributing pretrained model training to the current experiments.

Three research questions guide the analysis:

1) How does a 200-candidate Reader stage change DPR Multi answer-hit accuracy on full Natural Questions, CuratedTREC, and TriviaQA tests?
2) What quality and runtime changes occur when the Reader candidate budget increases from 100 to 200 on the tests with completed comparisons?
3) How much of the difference in absolute accuracy across datasets is associated with answer availability in the fixed corpus, and what remains unresolved about the larger TriviaQA gain?

The contribution is an empirical evaluation with explicit scope. First, we connect the official retriever, exact FAISS search, and the original Reader relevance head in a reproducible inference pipeline. Second, we report full-test lexical and dense baselines on five benchmarks and completed reranking results on three of them. Third, we distinguish answer-string corpus coverage from retrieval accuracy and derive a coverage-conditioned view of the same saved hit counts. Finally, a candidate-budget comparison exposes a case in which deeper answer recall improves while early ranking does not improve uniformly.

The work does not introduce a new neural architecture, a new loss, or a new indexing algorithm. Its value is in the measured behavior, transparent implementation choices, and the separation of corpus and ranking limitations. These distinctions are necessary when presenting a small-corpus reproduction as scientific evidence rather than treating it as a direct substitute for the original benchmark.

II. Related Work

A. Lexical and Dense Retrieval

BM25 ranks passages from lexical evidence with term-frequency saturation and document-length normalization [3]. It remains a useful reference because lexical overlap can identify specific entities and expressions that a dense model may rank poorly. Our lexical reference uses Apache Lucene,

with implementation details stated separately from those of the DPR paper.

DPR encodes a question and a passage independently with BERT-based encoders [1], [4]. Passage representations can be computed once and reused across questions. The independence of the two encoders enables efficient vector search, while a fixed passage vector must support many potential questions. In our experiments the released weights are held fixed; the original contrastive training objective explains the model's provenance rather than describing a training run performed here.

B. Vector Search

FAISS provides methods for exact and approximate similarity search [5]. HNSW organizes vectors in a graph to obtain approximate nearest neighbors [6]. The DPR paper reports an HNSW deployment for its large corpus. At 50,000 passages, exact inner-product search is feasible and avoids approximate-search errors relative to the DPR scoring function. Exactness of vector search is distinct from answer relevance: a correctly retrieved highest-scoring vector can still correspond to an unhelpful passage.

C. Neural Reranking and Readers

BERT-based reranking is an established way to score a query and a passage jointly [4], [7]. The original DPR system also provides an extractive reader with passage-selection and answer-span heads [2]. We reuse its passage-selection head as a reranker; we do not use an independently trained MS MARCO reranker and do not optimize a new reader. This distinction matters because our evaluation measures the ranking of answer-containing passages, not the exact-match accuracy of extracted answer spans.

Natural Questions and TriviaQA supply complementary question-answering benchmarks [8], [9]. We use the official DPR-formatted test resources, including the CuratedTREC, WebQuestions, and SQuAD resources distributed with that repository, rather than creating a random train–validation–test split of Wikipedia passages.

III. Methodology

A. Corpus Acquisition and Preprocessing

We sample the official DPR Wikipedia passage resource (`data.wikipedia_split.psgs_w100`). The recorded archive size is 4,694,541,059 bytes, and the manifest records 21,015,324 source passages. A streaming reservoir sampler selects 50,000 distinct passage IDs without replacement using seed 12345. Each source passage has the same inclusion probability. Sampling does not use test questions, answer strings, or known positive passages to construct an answer-enriched collection.

The sample is approximately 0.238% of the source collection. Its tab-separated records retain `id`, `text`, and `title`. Multiple passages may share an article title. No new passage segmentation is performed after sampling, and the official compressed source is retained separately. The corpus file used in the Kaggle reranking inputs and the local coverage measurement has the same SHA256 digest; the digest is recorded in the accompanying artifact manifest.

B. DPR Dual Encoder and Exact Search

Let q denote a question and p a passage. The DPR question and passage encoders produce independent 768-dimensional representations:

$$u = E_Q(q), \qquad v_p = E_P(p), \qquad s_D(q,p) = u^\top v_p. \quad (1)$$

The encoders use pretrained `bert-base-uncased` architectures and released DPR checkpoint weights. The retriever sequence limit is 256 tokens. Passage inputs use title and body text through the DPR tensorizer. Vectors are not unit-normalized for cosine similarity. Passage vectors are computed with the passage encoder corresponding to the selected retriever checkpoint and are associated with their passage IDs.

The original DPR training objective ranks a positive passage against explicit and in-batch negatives. For context, its per-question negative log-likelihood has the form

$$\mathcal{L}_i = -\log \frac{\exp s_D(q_i, p_i^+)}{\exp s_D(q_i, p_i^+) + \sum_{p^- \in \mathcal{N}_i} \exp s_D(q_i, p^-)}. \quad (2)$$

We do not recompute this loss or update encoder weights in the experiments reported here. In particular, test inference does not have a training batch size or training epoch count.

FAISS `IndexFlatIP` exhaustively scores the sampled passage vectors. The raw float32 matrix contains $50{,}000 \times 768 \times 4 = 153{,}600{,}000$ bytes, approximately 146.5 MiB, excluding ID mappings, model memory, and duplicate buffers. This is manageable at the selected corpus size. The published large-corpus DPR deployment uses HNSW with 512 stored neighbors, construction depth 200, and search depth 128 [1]. Both index wrappers are available in the original code. We do not report a paired HNSW–FlatIP experiment, so no measured accuracy increase is assigned to the index substitution itself.

C. Candidate Construction

For a question q, exact DPR search produces a candidate set

$$\mathcal{C}_K(q) = \mathrm{TopK}_{p \in \mathcal{P}}\, s_D(q,p). \quad (3)$$

The primary reranking experiment uses $K = 200$. Each returned passage ID retrieves the title and body from the fixed corpus. The Reader input is constructed as

$$\texttt{[CLS]}\ q\ \texttt{[SEP]}\ t_p\ \texttt{[SEP]}\ p. \quad (4)$$

The implementation appends body tokens to the question–title encoding, then truncates or pads the combined sequence to the Reader limit of 350 tokens. It does not unconditionally append another separator after the body. Token type IDs distinguish the question segment from title and passage; the attention mask excludes padding. Gold answer strings are not supplied to the Reader.

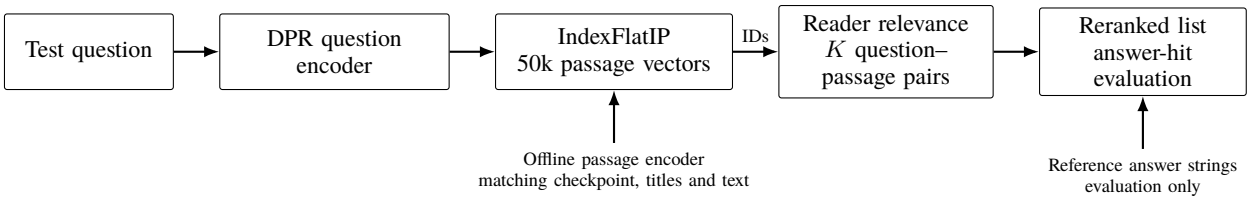


Fig. 1. Inference pipeline. Passage vectors are generated offline with the matching DPR checkpoint. Each Reader input contains one question and one candidate's title and text. Answers enter only the evaluation branch, after ranking.

D. Reader Relevance Reranking

Unlike the dual encoder, the Reader processes the concatenated question–passage input with one BERT encoder. Self-attention can connect tokens across the question, title, and passage. If $h_{\text{CLS}}(q, t_p, p)$ is the final hidden vector of the initial token, the original `qa_classifier` computes

$$s_R(q,p) = w^\top h_{\text{CLS}}(q, t_p, p) + b. \tag{5}$$

The reranked list sorts $\mathcal{C}_K(q)$ by s_R. We use raw relevance logits, not batchwise probabilities, and do not add the DPR score to the Reader score. The original Reader also computes start and end logits; these outputs are ignored by the reranking script.

We use the official `nq-trivia-hybrid` Reader checkpoint; its full resource identifier is recorded in the artifact manifest. The registry describes it as a reader trained using Trivia multi-hybrid retrieval results [2]. This provenance motivates a possible dataset-alignment explanation, but does not establish how much of a measured gain is caused by training-domain alignment.

Reader evaluation uses `torch.inference_mode`. A batch contains 16 candidate pairs, represented as $[16, 1, 350]$ before the Reader flattens its first two axes. DataParallel divides the first axis across two T4 GPUs. Each pair is scored independently; there is no cross-passage attention between the 200 candidates. No parameters are updated, and the existing DPR passage vectors are reused.

E. Lexical Reference

The BM25 reference is a standalone Java runner using Lucene 8.11.2, `EnglishAnalyzer`, and `BM25Similarity` with $k_1 = 0.9$ and $b = 0.4$. It indexes article title and passage body together, while answer matching uses body text only. This is a Lucene reference, not the exact software deployment of the original paper.

Previously saved score-fusion outputs are excluded because a review found inconsistent passage identifier formats between the lexical and dense branches. They cannot be treated as a validated hybrid baseline. No corrected fusion experiment is claimed here.

IV. Experimental Setup

A. Datasets and Evaluation Scope

The saved baseline evaluations cover all 28,219 test questions from the five official DPR-formatted benchmarks. Reader reranking is completed on all questions from NQ, CuratedTREC, and TriviaQA, totaling 15,617. We do not subsample or repeatedly redraw test questions for this report. WebQuestions and SQuAD remain first-stage references and are not presented as completed reranking experiments. Table I lists the denominators and corpus coverage.

TABLE I
Full test sizes and answer-string coverage in the 50k corpus. Coverage is not retrieval accuracy.

Test	Questions	Covered	Cov. (%)	Rerank
NQ	3,610	2,192	60.72	Yes
TriviaQA	11,313	8,233	72.77	Yes
WebQuestions	2,032	1,388	68.31	No
CuratedTREC	694	217	31.27	Yes
SQuAD 1.1	10,570	6,112	57.82	No

The DPR Multi checkpoint is the fixed first stage for the Reader comparisons. An additional official NQ-Single checkpoint is reported for context. It is a task-matched single-dataset model only for NQ; its other rows are NQ-to-target transfer evaluations, not reproductions of the original paper's task-specific Single models. No new checkpoint is selected using test accuracy. The $K = 100$ versus $K = 200$ comparison is exploratory: candidate budget has not been optimized on a held-out development set, and $K = 200$ is not asserted to be optimal.

B. Answer-Hit Metric

Let A_q contain the accepted answer strings of question q, and let $m(a, p)$ indicate that the normalized token sequence of answer a occurs contiguously in the body of passage p. For ranked list $R(q)$,

$$\text{Hit}@k = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\exists p \in R_k(q),\ a \in A_q : m(a,p)]\,. \tag{6}$$

We use Unicode NFD normalization, DPR SimpleTokenizer, and case-insensitive token-phrase matching. Empty normalized answer strings are excluded. The same convention is applied to first-stage and reranked outputs, and titles are excluded from answer matching. All displayed rates are recomputed from integer hit counts; percentage-point differences use unrounded rates.

This follows the answer-containing-passage retrieval convention. It is not an annotated relevance judgment and is not end-to-end QA Exact Match. A passage can contain the answer

string without supporting the required relation; alias lists and common short answers can also affect measured hits.

C. Coverage and Coverage-Conditioned Accuracy

Corpus coverage is measured over all 50,000 passage bodies with the same normalization and token matching:

$$\text{Cov} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\exists p \in \mathcal{P},\ a \in A_q : m(a, p)]\,. \tag{7}$$

The stored diagnostic scans normalized passage tokens with a multi-pattern token matcher. It does not use retrieval scores. Under this evaluator, Hit @$k \leq$ Cov for any retriever searching this corpus. Coverage is therefore an answer-string availability ceiling, not a score achieved by a model.

For questions whose answer string appears somewhere in the corpus, we additionally derive

$$\text{Hit @}k \mid \text{covered} = \frac{\text{number of hits at } k}{\text{number of covered questions}}. \tag{8}$$

This diagnostic changes the denominator. It must be reported alongside full-test accuracy, not substituted for it when comparing with a benchmark paper. The original full test remains the evaluation set; no questions are removed from the primary results.

D. Runtime and Reproducibility

The completed Reader runs use two Kaggle Tesla T4 GPUs, a global candidate batch of 16, and the FP32 inference path of the script. Retriever question batches are eight. Passage embeddings, corpus records, and the relevant test files are supplied as fixed inputs. The corpus copy used by reranking is byte-identical to the coverage corpus. Runtime in this report is the script's reranking-loop timer, which includes pair preparation, Reader scoring, and hit matching but excludes queueing, environment setup, model loading, initial question encoding, and FAISS search. First-stage timing from another execution environment is not used to calculate a purported end-to-end speed ratio.

There is one corpus seed and one completed run per reported configuration. The retained reranking JSON files contain aggregate counts rather than per-query transitions, so we report descriptive changes without paired significance tests, error bars, or claims of statistically established superiority.

V. Results

A. First-Stage References on Five Tests

Table II reports the full-test lexical and dense references. DPR Multi exceeds BM25 at both reported cutoffs on all five datasets in this sampled collection. The NQ-Single checkpoint is weaker than Multi on the three reranking targets, but its non-NQ values must be interpreted as transfer results. The Reader comparisons use Multi alone as the first stage; they do not depend on the excluded score-fusion implementation.

The sample changes answer availability substantially. Across all five tests, 18,142 of 28,219 questions have a matching answer string somewhere in the corpus, or 64.29%.

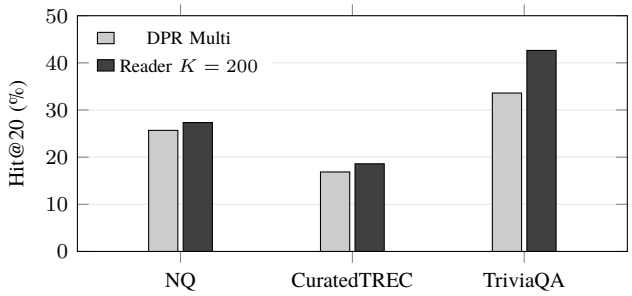


Fig. 2. Full-test answer-hit accuracy at rank 20 on the fixed 50k corpus. Bars describe completed runs; no uncertainty intervals were estimated.

On CuratedTREC the ceiling is only 31.27%, far below the attainable score on a collection with more answer evidence. The published full-corpus DPR values are consequently contextual references, not directly comparable targets for the present tables.

B. Reader Reranking on Three Full Tests

Table III gives all four cutoffs with hit counts. With $K = 200$, NQ rank-20 accuracy increases by 1.66 percentage points, from 927 to 987 hits. CuratedTREC increases by 1.73 points, from 117 to 129 hits. TriviaQA increases by 9.04 points, from 3,801 to 4,824 hits. The latter is a net increase of 1,023 successful questions at that cutoff, not a count of individually recovered queries with no regressions.

The rank-1 gain is also larger in absolute percentage points on TriviaQA: 8.64, compared with 0.91 on NQ and 4.32 on CuratedTREC. Relative gains tell a different story at rank 1: CuratedTREC improves by 88.24% and TriviaQA by 88.11% relative to their baseline hit counts. Hence, the assertion that TriviaQA improves the most depends on the stated measure. Figure 2 displays the primary rank-20 comparison.

At rank 100, the 200-candidate Reader adds 66 net hits on NQ and 399 on TriviaQA. CuratedTREC remains at 167 hits, or 24.06%. Equal totals do not establish that the successful questions are identical. The same aggregate count could result from improvements and regressions that cancel.

Pooling the three datasets by question count, rank-20 accuracy increases from 31.02% to 38.04%, a net 1,095 hits and 7.01 percentage points. TriviaQA contributes 72.44% of these questions. The unweighted mean of the three dataset accuracies increases from 25.38% to 29.52%, a smaller 4.14-point change. Both summaries are useful, but they answer different weighting questions and do not replace the dataset-level results.

C. Coverage Changes the Cross-Dataset Interpretation

Table IV separates availability from ranking. After Reader reranking, CuratedTREC has the lowest full-test rank-20 accuracy, 18.59%, yet retrieves an answer in the top 20 for 59.45% of its covered questions. This conditional rate is slightly above TriviaQA's 58.59% and above NQ's 45.03%. At rank 100, the conditional rates are 76.96%, 70.86%, and 64.51% for CuratedTREC, TriviaQA, and NQ, respectively.

TABLE II
FULL-TEST FIRST-STAGE REFERENCES. EACH CELL REPORTS HIT@20 / HIT@100 (%). NQ-SINGLE ON NON-NQ TESTS IS TRANSFER, NOT THE ORIGINAL PAPER'S TASK-SPECIFIC SINGLE SETTING.

Test	Lucene BM25	DPR NQ-Single	DPR Multi
NQ	18.86 / 29.72	22.85 / 35.57	25.68 / 37.34
TriviaQA	23.52 / 35.91	24.97 / 40.65	33.60 / 48.04
WebQuestions	26.97 / 38.48	28.94 / 44.00	36.27 / 48.18
CuratedTREC	12.82 / 17.58	13.11 / 21.47	16.86 / 24.06
SQuAD 1.1	16.15 / 27.59	18.66 / 31.75	21.42 / 34.37

TABLE III
PRIMARY RERANKING RESULTS ON THE SAME 50K CORPUS. CELLS CONTAIN ACCURACY IN PERCENT AND SUCCESSFUL-QUESTION COUNTS IN PARENTHESES.

Test	Method	Hit@1	Hit@5	Hit@20	Hit@100
NQ ($n = 3,610$)	DPR Multi	6.18 (223)	14.96 (540)	25.68 (927)	37.34 (1,348)
	Reader, $K = 200$	7.09 (256)	15.93 (575)	27.34 (987)	39.17 (1,414)
CuratedTREC ($n = 694$)	DPR Multi	4.90 (34)	11.82 (82)	16.86 (117)	24.06 (167)
	Reader, $K = 200$	9.22 (64)	14.12 (98)	18.59 (129)	24.06 (167)
TriviaQA ($n = 11,313$)	DPR Multi	9.81 (1,110)	20.77 (2,350)	33.60 (3,801)	48.04 (5,435)
	Reader, $K = 200$	18.46 (2,088)	31.32 (3,543)	42.64 (4,824)	51.57 (5,834)

TABLE IV
DERIVED COVERAGE-CONDITIONED ACCURACY (%). THE LAST THREE COLUMNS USE COVERED QUESTIONS AS DENOMINATOR; THEY ARE NOT FULL-TEST BENCHMARK SCORES.

Test	Corpus Cov.	DPR H@20| C	Reader H@20| C	Reader H@100| C
NQ	60.72	42.29	45.03	64.51
CuratedTREC	31.27	53.92	59.45	76.96
TriviaQA	72.77	46.17	58.59	70.86

TABLE V
COMPLETED CANDIDATE-BUDGET COMPARISONS. ACCURACY IS FULL-TEST PERCENT; TIME IS THE RERANKING LOOP IN SECONDS.

Test	K	Hit@20	Hit@100	Time (s)
NQ	100	27.67	37.34	4,488.34
NQ	200	27.34	39.17	8,554.71
CuratedTREC	100	18.88	24.06	832.97
CuratedTREC	200	18.59	24.06	1,636.29

This does not make CuratedTREC the best benchmark result: its full-test score is still low, and the conditional metric has a different denominator. It does show why low absolute accuracy cannot by itself establish that the Reader ranks CuratedTREC candidates worse. Uneven corpus answer availability explains an important part of the difference in attainable scores. It does not fully explain differences in the reranking gain, which also depends on which answer passages reach the candidate set and where the Reader places them.

D. Candidate Budget and Runtime

Table V compares the completed 100- and 200-candidate runs on NQ and CuratedTREC. Doubling candidates increases NQ rank-100 hits from 1,348 to 1,414, but NQ rank-20 hits decrease from 999 to 987. CuratedTREC rank-20 hits decrease from 131 to 129, while its rank-100 total is unchanged. Larger candidate sets therefore do not provide a monotonic improvement at every cutoff.

With exactly 100 unique candidates and a pure permutation reranker, the set of top-100 passages is unchanged, so rank-100 answer hits must remain invariant. With 200 candidates, passages originally ranked 101–200 can enter the new top 100. This mechanism permits deeper improvements but also lets high-scoring distractors displace previously useful candidates.

NQ reranking takes 4,488.34 seconds at $K = 100$ and 8,554.71 seconds at $K = 200$, a 1.91-fold increase. CuratedTREC takes 832.97 and 1,636.29 seconds, a 1.96-fold increase. The completed TriviaQA $K = 200$ run takes 28,197.25 seconds. Its stopped $K = 100$ run has no full-test final metrics and is not used as a completed comparison.

The 200-candidate runs evaluate 722,000 NQ pairs, 138,800 CuratedTREC pairs, and 2,262,600 TriviaQA pairs: 3,123,400 in total. Their recorded loop times sum to 38,388.25 seconds, approximately 10 hours 40 minutes across separate runs. Mean loop times per question are 2.37, 2.36, and 2.49 seconds. The close NQ and CuratedTREC means indicate that their different total runtimes largely reflect the number of evaluated pairs, rather than establishing a dataset-specific acceleration.

VI. DISCUSSION AND LIMITATIONS

A. Why Is the TriviaQA Gain Larger?

Two observations support plausible explanations. First, the registry associates the selected Reader release with Trivia multi-hybrid training outputs. We use that same checkpoint on all three targets without target-specific fine-tuning. Distributional alignment may help its relevance head on TriviaQA,

but a matched-budget Reader comparison would be needed to estimate the causal contribution of this factor.

Second, the initial DPR candidate pool already offers more answer-bearing opportunities on TriviaQA. Baseline rank-100 accuracy is 48.04% on TriviaQA, versus 37.34% on NQ and 24.06% on CuratedTREC. The difference between baseline rank-100 and rank-20 hits is 1,634 questions on TriviaQA, 421 on NQ, and 50 on CuratedTREC, corresponding to 14.44%, 11.66%, and 7.20% of the respective tests. These questions have at least one answer passage in ranks 21–100 and could benefit from promotion. These are candidate-pool observations, not measurements of Reader transitions or of all candidates at $K = 200$.

Joint question–passage encoding can exploit token-level relations that a fixed vector comparison ranks imperfectly. For instance, a passage can mention the correct entity while answering a different relation. The Reader's question-conditioned hidden state can help discriminate such cases. We have not retained per-query relevance scores or inspected attention maps, so illustrative explanations of these interactions are not offered as measured model internals.

B. Limits on Attribution

The main comparison combines a Reader stage with a 200-candidate budget. It establishes the observed performance of that pipeline relative to the baseline, not a complete decomposition of every improvement. Candidate-budget effects are partially observable on NQ and CuratedTREC, but no completed full-test TriviaQA $K = 100$ run exists. Exact IndexFlatIP is used before and after reranking; the reported gain is not a measured HNSW-to-FlatIP gain.

Single-seed corpus sampling is another limitation. Removing corpus material changes both the answer-bearing evidence and the distractor distribution. The sample is uniform at passage level, not balanced by question topic, answer type, or dataset. Coverage uses test answers only as a retrospective diagnostic; using those answers to redesign the corpus and then reporting the same test as an unbiased benchmark would change the protocol.

The answer-string metric can count unsupported mentions and is sensitive to answer aliases. The Reader sees at most 350 combined input tokens, whereas the evaluator searches the stored passage body; truncation can therefore hide evidence from the Reader that remains visible to the metric. No answer-span Exact Match result is available, and no claim is made about the factual correctness of generated or extracted answers.

C. Reproducibility Scope and Next Steps

The artifact package contains the LaTeX sources, bibliography, machine-readable result summaries, generated tables, a derivation script, and the corpus identity information needed to trace the reported numbers. The inference implementation remains in the accompanying DPR project. Released checkpoints and large datasets are not embedded in the paper package. Script paths and resource identifiers are provided instead. This is sufficient to audit the arithmetic without treating it as a packaged replacement for the GPU execution environment.

A stronger next evaluation would keep the corpus fixed, save per-query candidate IDs and relevance scores, choose the candidate budget on development data, and perform one final test evaluation. Repeating the corpus sampling across seeds would quantify sampling sensitivity. A paired HNSW–FlatIP comparison could separate vector-search effects from relevance reranking. Comparing Reader checkpoints with controlled training provenance would test the dataset-alignment explanation. These are proposed extensions, not completed results in this paper.

VII. Conclusion

On a fixed 50,000-passage Wikipedia sample, the official DPR Reader improves DPR Multi rank-20 answer-hit accuracy on all three completed full-test reranking benchmarks. Gains range from 1.66 percentage points on NQ to 9.04 on TriviaQA. The benefit is accompanied by substantial pair-scoring cost, and increasing the candidate budget from 100 to 200 does not improve every cutoff. Coverage-conditioned analysis further shows that CuratedTREC's low absolute accuracy coexists with a competitive retrieval rate among questions whose answer strings are present. Reduced-corpus studies should therefore report corpus availability, ranking quality, and computation separately. The present results characterize an inference pipeline using released models; they do not establish a new architecture, a full training reproduction, or scores directly comparable with the original 21-million-passage benchmark.

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